

Does Scale Close the Script Gap? Evaluating LLMs on Non-Latin Languages Using XQuAD

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Abstract

Large language models (LLMs) are predominantly trained on English-centric corpora, raising concerns about their performance on non-Latin scripts such as Hindi (Devanagari) and Thai. In this paper, we evaluate two LLaMA models of different scales – LLaMA 3.1 8B and LLaMA 3.3 70B – on the XQuAD benchmark across English, Hindi, and Thai using identical questions and contexts. Our results reveal that smaller models exhibit a consistent English-bias of 4–5% on non-Latin scripts, while scaling to 70B parameters not only eliminates this gap but reverses it, with Hindi and Thai outperforming English by 1–2%. Error analysis on the 8B model identifies number confusion, entity selection errors, and negation misunderstanding as primary failure modes in Hindi. Our findings suggest that cross-lingual transfer improves superlinearly with scale, with implications for the design of efficient multilingual models such as encoder-injection approaches.

1 Introduction

Despite rapid advances in large language models, performance disparities across languages remain a critical concern, particularly for non-Latin scripts. Languages such as Hindi (written in Devanagari script) and Thai face compounded challenges: tokenizer over-fragmentation, limited pretraining data, and reduced representation in instruction tuning datasets [Gupta & Agarwal, 2025]. While prior work has demonstrated cross-lingual transfer capabilities in multilingual models [Artetxe et al., 2020], the relationship between model scale and cross-lingual performance on non-Latin scripts remains underexplored.

In this work, we ask two questions: (1) Do LLMs exhibit a systematic English-bias on non-Latin scripts? (2) Does scaling the model reduce or eliminate this bias? We evaluate LLaMA 3.1

8B and LLaMA 3.3 70B on the XQuAD benchmark [Artetxe et al., 2020] across English, Hindi, and Thai – three typologically distinct scripts – using extractive question answering as our evaluation task.

2 Related Work

Cross-lingual QA. XQuAD [Artetxe et al., 2020] provides a gold-standard cross-lingual evaluation benchmark with identical questions translated across 12 languages, enabling controlled comparison across scripts. Prior work using mBERT and XLM-R [Conneau et al., 2020] demonstrated strong cross-lingual transfer for Latin-script languages but noted consistent degradation for non-Latin scripts.

Low-resource multilingual LLMs. Recent work has proposed architectural solutions to cross-lingual alignment. LLINK [Gupta & Agarwal, 2025] treats low-resource languages as modalities and injects multilingual encoder embeddings into a frozen decoder via a contrastive projector, achieving strong cross-lingual performance without full retraining. Our findings complement this line of work by demonstrating that at sufficient scale, vanilla LLMs may close the script gap without architectural modifications.

3 Experimental Setup

Dataset. We use XQuAD [Artetxe et al., 2020], which contains 1,190 question-answer pairs translated into 12 languages from the SQuAD v1.1 development set. We evaluate on 100 examples per language for English (`xquad.en`), Hindi (`xquad.hi`), and Thai (`xquad.th`), ensuring identical questions across all three languages.

Models. We evaluate two models from the LLaMA family via the Groq API:

- **LLaMA 3.1 8B** (llama-3.1-8b-instant): a compact, efficient model representing the small-scale setting.
- **LLaMA 3.3 70B** (llama-3.3-70b-versatile): a large model representing the high-scale setting.

Evaluation. We use zero-shot prompting with a fixed template instructing the model to answer in a short phrase. Accuracy is measured as substring match between the predicted answer and any gold answer string, case-insensitive.

4 Results

Model	English	Hindi	Thai
LLaMA 3.1 8B	91%	86%	87%
LLaMA 3.3 70B	93%	94%	95%

Table 1: Zero-shot accuracy on XQuAD across three scripts.

As shown in Table 1, LLaMA 3.1 8B exhibits a consistent English-bias of 4–5% on non-Latin scripts. Strikingly, LLaMA 3.3 70B not only eliminates this gap but reverses it: Hindi (94%) and Thai (95%) both outperform English (93%). This suggests that cross-lingual transfer improves superlinearly with model scale.

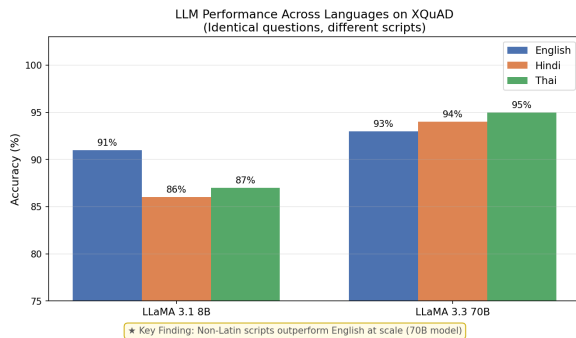


Figure 1: Accuracy comparison across languages and model scales.

5 Error Analysis

We manually analyzed the 14 failure cases of LLaMA 3.1 8B on Hindi and identified four error categories:

- **Number confusion** (2 cases): The model predicted Hindi word numerals (e.g., *chaar* meaning “four”) when the correct answer was a digit (24).
- **Wrong entity selection** (1 case): The model selected a plausible but incorrect named entity from the context.
- **Negation misunderstanding** (1 case): The model failed to correctly interpret negation in Hindi, returning the wrong subject.
- **Off-by-one numeric errors** (1 case): The model predicted a number one off from the correct answer (38 vs. 39).

These error types align with known weaknesses of smaller LLMs on non-Latin scripts: tokenizer over-fragmentation of Hindi numerals and reduced sensitivity to grammatical markers such as negation.

6 Discussion

Our results have two key implications. First, model scale is a strong predictor of cross-lingual fairness – small models should not be deployed in multilingual settings without explicit cross-lingual alignment. Second, the reversal of the English-bias at 70B scale suggests that LLaMA 3.3’s training data may contain substantial non-Latin script content, enabling stronger cross-lingual transfer than its smaller counterpart.

These findings motivate continued work on efficient cross-lingual alignment for smaller models, such as encoder-injection approaches [Gupta & Agarwal, 2025], which aim to achieve large-model cross-lingual performance at small-model computational cost.

7 Conclusion

We evaluated LLaMA 3.1 8B and LLaMA 3.3 70B on English, Hindi, and Thai using the XQuAD benchmark. Our findings demonstrate a consistent English-bias in smaller models that is eliminated and reversed at larger scale. Error analysis reveals number confusion and negation misunderstanding as primary failure modes in Hindi for smaller models. Future work will extend this analysis to few-shot settings, additional Indic languages, and encoder-injection models.

References

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